

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Quiz - 3
 Duration: 20 minutes

Semester: Summer 2026
 Full Marks: 20

CSE 340: Computer Architecture

Name: <u>Solution</u>	ID:	Section:
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1. Suppose we run the following instruction set in a **pipelined datapath**. Find the number of Data hazards in each line. [1 x 11]

Instruction Set	Number of Data Hazard(s)
add <u>x5</u> , x6, x7	<u>0</u>
sub x5, <u>x5</u> , x9	<u>1</u>
ld x10, 0(<u>x8</u>)	<u>0</u>
and x11, <u>x10</u> , <u>x5</u>	<u>2</u>
or x12, <u>x11</u> , x8	<u>1</u>
sd <u>x12</u> , 8(<u>x10</u>)	<u>1</u>
addi x13, <u>x12</u> , 4	<u>1</u>
xor x14, <u>x13</u> , <u>x11</u>	<u>1</u>
sll x15, <u>x14</u> , x12	<u>1</u>
addi x12, <u>x15</u> , <u>x14</u>	<u>2</u>
sd x12, 16(<u>x15</u>)	<u>2</u>

2. Analyze the tables given below. Table 1 assumes that individual stages of a **single cycle datapath** have the following latencies; some values are missing. Table 2 includes the total time required to execute some of the instructions using the given datapath. Using the given data, find the time required for ID, MEM, and WB stages.

[2 + 2 + 2 + 3]

Table - 1:

IF = 150ps	ID = <u>120</u>	EX = 180ps	MEM = <u>250</u>	WB = <u>80</u>
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Table - 2:

Instruction	Time (PS)
Add X ₇ , X ₁₈ , X ₂₁	530
BNE X ₂₁ , X ₈ , L1	450
SD X ₂₉ , 8(X ₁₁)	700
LD X ₂₆ , 0(X ₂₅)	780

$$F + D + E + W = 530 \Rightarrow W = (530 - 150 - 120 - 180) =$$

$$F + D + E = 450 \Rightarrow D = (450 - 150 - 180) = 120$$

$$F + D + E + M = 700 \Rightarrow M = (700 - 150 - 120 - 180) = 250$$

What would be the clock period of the system? Answer:

~~780~~ 780 ps